

# Your Index Is Fine, Your Query Encoder Is Not: Query Distillation over Frozen Indexes

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October 2026

## Abstract

A dense index encodes each document once; its query encoder runs on every uncached search. Replacing only the query encoder preserves stored vectors, graphs, and quantized codes, and can cut median CPU query-encoding time from 31.6 ms to 0.044 ms. How much quality survives, and what the replacement costs at search time, depends on the index it inherits.

A student is a smaller query encoder trained to reproduce the original model’s query vectors. The study fits token tables that average learned token vectors and transformer students that project features from a frozen transformer. Among ten registered embedding models, the best retriever produces the worst table student. Across 26 models, dimensionality and measured retrieval score jointly predict student quality, with prospective support for the transformer student. On unchanged HNSW graphs, fitted tables need a median two to four times the search breadth to recover the same fraction of their own exact neighbors. This deficit recurs in a jointly trained lookup model; the index’s own queries flag where it is largest.

Stella, a 400-million-parameter English embedding model, supports two separately trained students without document re-encoding: the 34.5-million-parameter transformer Nano and token table Zero. They retain 90.5% and 81.4% of Stella’s BEIR-15 nDCG@10, respectively. In an exploratory selection on one binary-quantized collection, encoding plus search falls from 21.3 ms to 2.48 ms and 1.11 ms, respectively, at lower relevance. Nano’s final optimization took 57.3 A100 hours. The shared index lets each request choose its query encoder.

## 1. Introduction

Query and document encoders serve different computational roles. Document embeddings are stored and reused across searches, allowing an expensive document representation to be amortized. Every uncached query incurs its own encoding cost. This asymmetry raises the question of choosing the query encoder independently of the stored document representation. Query-side distillation builds a cheaper encoder whose vectors remain compatible with the documents [QED, EmbedDistill, LEAF], separating the representation retained in the index from the computation used to search it.

The stored document vectors support the graph, quantized representations, and downstream uses of those vectors. A compatible replacement query encoder can search these assets without changing them. Changing document encoders requires corpus re-embedding and index rebuilding; maintaining availability during migration can require parallel indexes and ingestion pipelines, extra infrastructure load, relevance validation, and rollback [Index migration].

The study asks which properties of an existing document representation predict student retrieval quality, and how replacing the query encoder changes the graph-search effort required. It fits two kinds of student across 26 embedding models: a table student that averages learned token vectors, and a transformer student that uses a frozen transformer with a learned projection. Each model retains its own document index.

**Constella** makes the choice of query encoder concrete. It consists of two query encoders aligned to the 1024-dimensional document embeddings of **Stella**, the pretrained English model `stella_en_400M_v5` [Stella]. **Nano** is a separately trained 34.5-million-parameter transformer, and **Zero** is a trained token-vector table. Together with Stella’s own query encoder, they provide three query-compute budgets over identical document vectors. A request can select an encoder without changing the document representation. Constella demonstrates the resulting choices with separately trained encoders, accounting for their construction costs and recurring encoding and search costs.

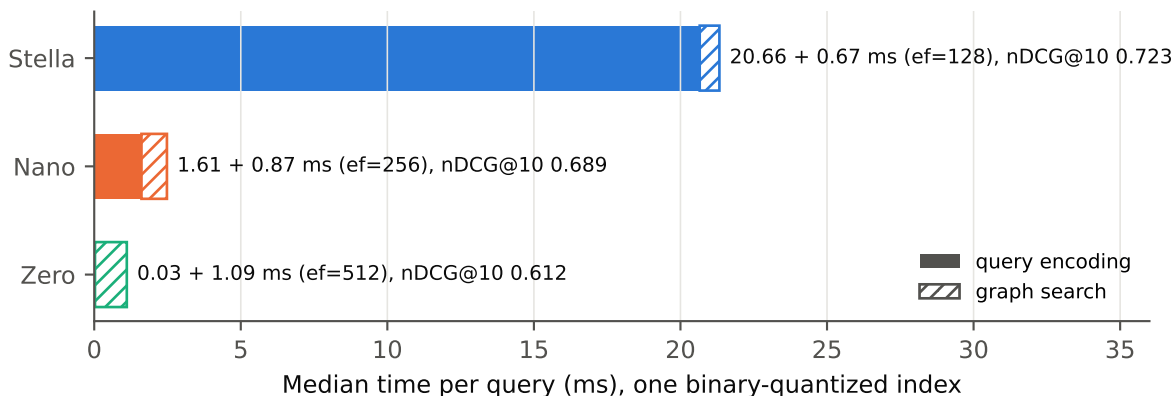


Figure 1: Where query time goes when three encoders search one unchanged index. Solid bars are query encoding and hatched bars are graph search, for each encoder at its fastest setting within 1% of its own exact score on one binary-quantized, positive-preserving one-million-passage MS MARCO diagnostic. Stella’s time is almost all encoding; Zero’s is almost all search, at four times Stella’s *ef*. The values are medians on the diagnostic’s own queries, with encoding and search measured separately, from an exploratory selection of the registered sweep at unequal relevance.

The study establishes three findings:

- **The best retriever can produce the worst cheap query encoder.** In the ten-model registered comparison, the best-retrieving gte-large produces the worst table and Stella the best, both at 1024 dimensions. The exploratory analysis finds that dimensionality and measured benchmark score jointly predict fitted-student quality before a candidate is fitted. A prospective test with predictions committed before fitting supports this for the transformer student (+0.86 [+0.33, +1.00]); the table result is uncertain.
- **Replacement queries need a wider search over the unchanged graph.** In the exploratory comparison, fitted tables need a median two to four times the graph-search breadth (*ef*) to match the model’s own exact-neighbor recovery. A distance ratio computed from the model’s own queries predicts recovery deficits at fixed effort without relevance labels or a fitted student. The deficit also recurs in LightRetriever’s jointly trained lookup query encoder.
- **An existing index can serve three query budgets without re-ingestion.** Nano and Zero were added to Stella’s populated collections without document re-encoding or collection rebuilding. On one binary-quantized million-passage MS MARCO diagnostic, encoding plus search takes 21.3 ms for Stella, 2.48 ms for Nano, and 1.11 ms for Zero, at nDCG@10 of 0.7226, 0.6893, and 0.6116, respectively, an exploratory selection at each encoder’s fastest setting within 1% of its own exact score (Figure 1). Nano’s final optimization took 57.3 A100 hours; the fitted screens take minutes.

## 2. Related work

Query-side distillation over a frozen document encoder is established. QED aligns a compact query encoder with an existing dense retriever [QED]; EmbedDistill studies geometric distillation against frozen document representations [EmbedDistill]; and LEAF provides teacher-aligned compact representations [LEAF]. pyNIFE fits a static token table against a frozen teacher [pyNIFE]. Model2Vec and static sentence embeddings average token vectors [Model2Vec, StaticEmb], placing Zero within a broader family of static representations. LightRetriever trains a lookup query encoder jointly with a document tower [LightRetriever]. Our contribution is a common-recipe comparison across embedding models and a study of the search consequences of substituting queries over unchanged graphs. QED reports 92.5% retention of a dense retriever’s BEIR score with a two-layer student, and LEAF reports 97.7% retention at 4.7-fold compression [QED, LEAF]. LightRetriever reports about 95% retention for its jointly trained lookup query encoder, which requires its own document index, compared with Zero’s 81.4% over frozen Stella here; the different models and evaluation protocols leave the quality cost of freezing the document index unresolved [LightRetriever].

Stronger teachers can produce weaker students in general distillation [Cho and Hariharan 2019] and dense retrieval [PROD]. Those studies discuss capacity gaps between teacher and student. We test embedding dimensionality as a predictive property across embedding models, alongside the measured relevance of the model’s own query encoder.

Graph-based approximate nearest-neighbor search is sensitive to query placement relative to indexed vectors [HNSW]. Out-of-distribution queries can require more work, and query-aware graph construction can mitigate difficult query distributions [OOD-DiskANN, RoarGraph]. Here the graph stays fixed while the query encoder changes.

Per-query retrieval-strategy selection and query performance prediction also have substantial prior work [Arabzadeh et al., QPP]. A compatible family permits selection, but compatibility alone supplies no rule for allocating compute according to relevance.

## 3. Experimental setup

### 3.1 Embedding models, students, and data

The study uses 26 public English embedding models sharing the 30,522-entry BERT WordPiece vocabulary. Ten embedding models were registered before results, and 16 were added as an exploratory expansion. The model list spans nine related model families and embedding dimensionalities of 384, 768, and 1024. Each model’s own query encoder searches its own frozen document vectors. Appendix C retains the complete scores and model identities.

We fit two representations for the students by ridge regression to each model’s own query vectors. The **table student** assigns a vector to each token, mean-pools the token occurrences, and L2-normalizes the result. Its regression penalty anchors the rows to vectors obtained by encoding individual tokens through the original model. The **transformer student** uses contextual features from a frozen bge-small transformer [BGE]: attention-masked mean states from layers 12, 8, and 4 are concatenated into 1,152 features. A fitted projection with bias maps those features into the stored document representation, followed by normalization. Only the projection is learned, so construction is inexpensive while inference still runs the backbone. The shared-vocabulary requirement applies to the table comparison. The transformer student obtains its features from bge-small.

Both students use the same cleaned 337,981 fit queries and each original model’s published query prompt for targets. Penalties are selected by exact retrieval on the CQADupStack physics and programmers development forums, then frozen before public scoring. Quality is normalized discounted cumulative gain at rank 10, **nDCG@10**, under exact search on NFCorpus, SCIDOCS, SciFact, TREC-COVID, FiQA, and ArguAna [BEIR]. The primary outcome is the unweighted mean over these six datasets. A sensitivity analysis uses the first four datasets, for which Stella discloses no training exposure.

The fitted students provide a controlled construction comparison. Released Zero and Nano use additional training beyond these fitted students and different training pools. Zero learns token vectors and scalar weights

through vector alignment, teacher-score matching, and contrastive ranking against frozen Stella document vectors. Its served table uses square-root token-count weights and int8 rows. Nano starts from bge-small, concatenates the same three layers, and learns a projection into Stella’s document embeddings; backbone and projection are then trained jointly against Stella targets. Training alternates three query batches with one document batch. Stella remains the document encoder for retrieval. We report the fitted students and the released models as separate experiments, with reproduction details in Appendix B.

### 3.2 Search and evidence status

Approximate search uses Qdrant 1.19.1, a vector search engine, with uncompressed vectors, cosine distance, and hierarchical navigable small-world (**HNSW**) graphs [Qdrant 1.19.1, HNSW]. The search workloads are FiQA, SCIDOCs, and TREC-COVID. Two graph builds per model differ in seeded document insertion order. The graph parameters stay fixed while each model’s own query encoder and its students search with varying **ef**, the parameter controlling graph-search breadth.

We measure **relevance loss**, the fractional reduction from an encoder’s own exact nDCG@10, and **exact-neighbor recovery**, the fraction of its exact top-10 recovered by graph search. Each encoder has its own reference. Effort multipliers compare the smallest student **ef** reaching the loss or recovery of the model’s own query encoder at **ef**=64, divided by 64 and averaged over builds. A target not reached by the largest setting is censored, with a multiplier of at least eight. Appendix B specifies ties, parity checks, and exclusions.

“Registered” identifies evaluations whose decision rules preceded observations. “Exploratory” identifies subsequent analyses. The quality and search screens across embedding models are exploratory, with their analyses written before scoring. The conditional-quality hypothesis was developed after seeing the original 26 embedding models; its prospective test committed predictions by hash before fitting new students.

## 4. Quality: what an existing index predicts

An existing index fixes the document representation, while its dimensionality and the model’s measured baseline retrieval are available before any student is fitted. We initially hypothesized that stronger baseline retrieval would predict stronger fitted students, so that the best-retrieving embedding model would yield the best student. The registered comparison reverses that selection rule: gte-large-en-v1.5 scores 0.5970 but yields the weakest registered table at 0.2455; Stella scores 0.5745 and yields the strongest registered table at 0.3974. Across all 26 embedding models (Figure 2), retrieval scores of the models’ own query encoders and their students have rank correlation +0.09 for both representations, with 95% intervals from resampling the models of  $[-0.39, +0.52]$  for the table and  $[-0.36, +0.51]$  for the transformer student.

The competing hypothesis uses both properties available for the existing index: at a given embedding size, higher baseline retrieval scores should favor stronger students, while higher dimensionality should predict weaker students after accounting for baseline retrieval quality. We model absolute student nDCG@10 from the model’s own retrieval quality and log2 dimensionality (Table 1). Standardized coefficients express changes in standard deviations of student retrieval scores per predictor standard deviation. Intervals use 10,000 resamples of the models; held-out prediction fits on eight families and predicts the ninth.

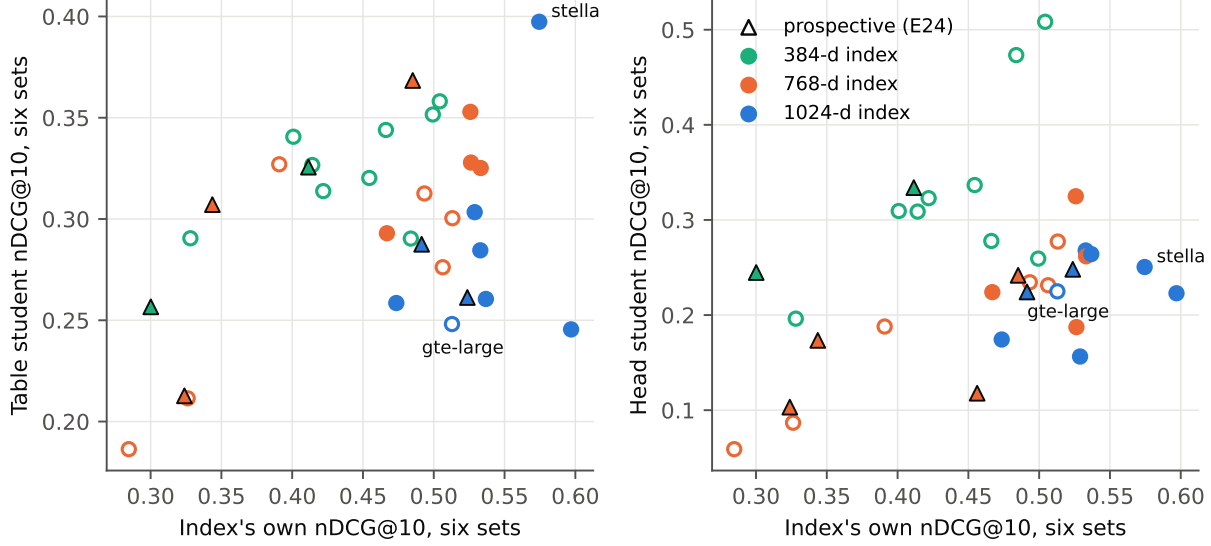


Figure 2: Student quality against the quality of the model’s own query encoder for table students (left) and transformer students (right), colored by dimensionality. Filled points are registered embedding models; triangles are prospective embedding models, eight for the transformer student and seven for the table.

**Table 1.** Conditional prediction of student nDCG@10 on the six datasets across 26 embedding models (exploratory, analysis pre-specified). Coefficients have 95% intervals;  $R^2$  is in-sample fit, and held-out  $\rho$  is Spearman correlation across family-held-out predictions.

| Student             | Quality coefficient  | Dimensions coefficient | $R^2$ | Held-out $\rho$ |
|---------------------|----------------------|------------------------|-------|-----------------|
| Transformer student | +0.69 [+0.40, +0.97] | −0.83 [−1.14, −0.60]   | 0.70  | +0.69           |
| Table student       | +0.63 [+0.06, +0.89] | −0.64 [−0.95, −0.32]   | 0.48  | +0.40           |

The model’s own retrieval quality and dimensionality correlate at +0.61. Their opposing associations with student quality explain why the pooled comparison obscures the positive quality signal. Quality-only models have  $R^2$  of 0.12 for the transformer student and 0.14 for the table, with held-out correlations −0.17 and −0.15. Adding dimensionality improves both. On the sensitivity outcome for four datasets, the dimensionality association persists, while the table’s interval for the coefficient of the model’s own retrieval quality slightly crosses zero. The result is predictive: the model list does not separate dimensionality causally from family and other model properties.

Because the hypothesis was formed after all 26 embedding models had been seen, we tested it prospectively: the test fixed eight further encoders, committed predictions by hash, then fitted and scored their students. One table fit stopped at its convergence gate, leaving seven tables and eight transformer students.

The prospective predictions rank the transformer students at +0.86 [+0.33, +1.00] and tables at +0.71 [−0.12, +1.00], against +0.26 and +0.39 for the model’s own retrieval quality alone. Mean absolute score error is 0.039 for the transformer student and 0.032 for the table. The transformer student result supports positive prospective ranking; the table interval leaves its result uncertain. Appendix C gives every actual and predicted score.

The prospective result supports predicting student rankings from properties of the original model. Selecting a single student is a further test: the following comparison measures how much retrieval quality each selection

rule loses relative to the best available student. On the original model list, the conditional model’s top choices, each predicted by a fit on the other 25 embedding models, lose 0.249 nDCG@10 for the transformer student and 0.107 for the table relative to the best available students. Selecting by directly measured development retrieval loses 0.035 and 0.000; on the prospective candidates, both rules select the best transformer student and a table 0.043 below the best.

Direct evaluation selected stronger students on the original model list, and its construction cost was modest: table fits took four to seven minutes per registered embedding model on an A100, including query encoding except for cached Stella targets. Direct evaluation also tests the relevant outcome: two e5 tables match their models’ own query vectors at mean cosine 0.90 and 0.89, against Stella’s 0.78, yet retrieve worse. Better vector agreement does not certify better ranking against the stored documents.

## 5. Search cost: what query replacement changes

Exact retrieval establishes the student representation’s quality. Approximate search adds another possible loss over the same document vectors. We hypothesized that replacing the model’s own queries would require more graph effort to recover the same fraction of each encoder’s exact neighbors. This exploratory screen holds the graphs fixed between query encoders and measures 25 models after the parity exclusion described in Appendix B.

At **ef=64**, table students recover fewer exact neighbors (Figure 3) for all 25 models on FiQA and TREC-COVID, and in 17 of 25 on SCIDOCS. Mean recovery gaps between students and the models’ own query encoders are  $-2.9$ ,  $-3.9$ , and  $-0.8$  percentage points, respectively. Median recovery-effort multipliers are four on FiQA and TREC-COVID and two on SCIDOCS. The transformer students also have deficits for 24 of 25 models on FiQA and 23 of 25 on TREC-COVID. Query substitution affects graph search beyond the lookup representation alone.

Relevance gives a less uniform answer. Matching the relative nDCG loss of the model’s own query encoder requires greater effort for all 25 models on FiQA, with a median multiplier of four among reached targets and four models censored in at least one build. The multiplier exceeds one for 17 of 25 models on SCIDOCS but only 12 of 25 on TREC-COVID. On TREC-COVID’s 50 queries, the model’s own query encoder loses only 0.2% of nDCG at the reference setting, making its threshold sensitive to very small differences. The recurring neighbor-recovery deficit therefore has corpus-dependent relevance consequences.

Corpus size contributes to the variation. An exploratory, pre-specified control subsamples FiQA to SCIDOCS’s 25,657 documents while retaining judged positives and rebuilding two graphs per model. The recovery gap moves from  $-2.9$  to  $-1.2$  points, a paired change of  $+1.8$  [ $+1.3$ ,  $+2.3$ ] across embedding models. The recovery deficit remains 0.4 percentage points larger than on SCIDOCS; the interval for the signed difference is  $[-0.9, -0.1]$ . Smaller corpus size reproduces much of the attenuation, with a remaining corpus difference.

Query placement also supplies a predictor. Before computing features, we specified the ratio of query-to-nearest-document distance to typical nearest-neighbor distance among documents, alongside other geometry measurements. Across 75 combinations of embedding model and workload, the ratio from the models’ own query vectors correlates with the recovery gap at  $-0.64$  [ $-0.75, -0.48$ ]. Its family-held-out predictions correlate at  $+0.64$  with mean absolute error 1.26 percentage points, versus 2.22 for the training-fold mean. Embedding models whose own query vectors sit farther from documents tend to incur larger replacement deficits.

The ratio for the model’s own queries marks the tendency toward larger recovery gaps, and its cause stays open: the difference in this ratio between the student and the model’s own queries predicts poorly on held-out families. Transfer across workloads is weaker than transfer across families: the model with all features and covariates, trained on two workloads, ranks the third at  $+0.50$ , but its error is 2.55 points against 2.33 for the fold-mean baseline.

We then test recurrence beyond students fitted to a frozen teacher. In an exploratory experiment specified before encoding, LightRetriever’s released Qwen2.5-1.5B model supplies both lookup and full query encoders over its own document vectors [LightRetriever]. Under its primary web-search prompt, lookup-minus-full

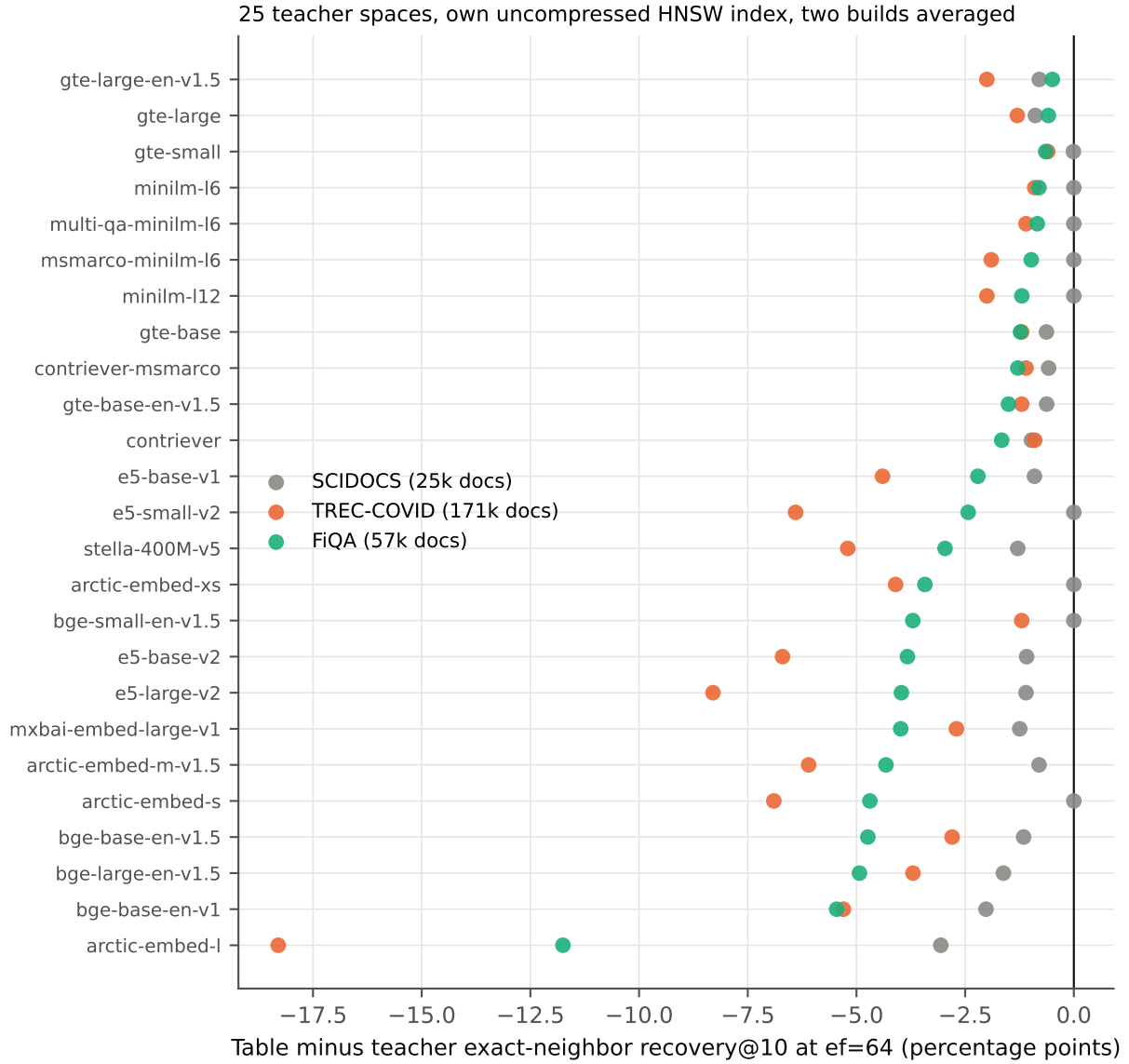


Figure 3: Student exact-neighbor recovery minus recovery with the model’s own query encoder at **ef=64**, by embedding model and workload, averaged over two builds. Negative values mean the table student recovers fewer of its own exact neighbors.

recovery gaps are  $-3.6$   $[-4.3, -2.9]$  points on FiQA and  $-1.7$   $[-2.1, -1.4]$  on SCIDOCS. Matching the full query encoder’s recovery requires four times the **ef** on both corpora. Relative relevance-loss effort agrees on FiQA; on SCIDOCS, both encoders lose under 0.1% of nDCG at the reference, making that comparison sensitive to small changes.

The deficit thus recurs in a jointly trained lookup model as well as our closed-form students. Search-effort multipliers measure graph breadth. Settings for each encoder recover the measured targets where the sweep reaches them; relevance and recovery remain distinct criteria for that adjustment.

## 6. One index, several query budgets

Constella turns query-side separation into a family: Stella, Nano, and Zero search the same populated document collections, while providing different relevance and compute budgets. Nano and Zero were added without document re-encoding or collection rebuilding between encoder batches, and all three encoders were verified against those same populated collections. The family preserves the stored representation across these choices, so the operating policy can choose an encoder for each request.

The registered descriptive BEIR-15 evaluation measures exact quality. Encoding is measured separately on an Apple M5 Pro CPU with four threads, batch one, and 100 real queries of five to 12 words. Table 2 places these two measurements side by side; its asset sizes describe encoder files, not resident memory.

**Table 2.** Constella over one frozen Stella document representation. Quality is exact BEIR-15 macro nDCG@10; retention is the ratio to Stella. Encoding medians follow the registered medium-query CPU protocol.

| Query encoder | nDCG@10 | Retention | Encode p50 | Asset size |
|---------------|---------|-----------|------------|------------|
| Stella        | 0.5614  | reference | 31.6 ms    | 1669.6 MiB |
| Nano          | 0.5081  | 90.5%     | 2.25 ms    | 132.3 MiB  |
| Zero          | 0.4572  | 81.4%     | 0.044 ms   | 90.1 MiB   |

Zero’s sum of token vectors is insensitive to word order and context. Its retention of Stella’s score is 0.67 on TREC-COVID and FiQA, compared with 0.95 on Quora (Figure A1). Nano’s retention ranges from 0.85 to 0.99 except on FEVER, which was in Zero’s training pool and excluded from Nano’s. The choice between them therefore depends on the type of query as well as on the compute budget.

A lexical branch supplies another compatible route. In the descriptive BEIR-15 evaluation, fusion with BM25 raises Zero’s score to 0.4933, or 87.9% of Stella’s score. The effect varies by workload; fusion lowers Nano’s score on MS MARCO. This route adds a sparse index and branch search, using distribution-based score fusion over 100 candidates per branch [DBSF].

On the uncompressed million-passages collection, additional graph search consumes much of Zero’s encoding advantage over Nano (Figure 4). On the one-million-passages MS MARCO diagnostic, with 6,980 development queries, uncompressed search reaches within 1% of each encoder’s own exact nDCG at **ef**=128 for Stella, 256 for Nano, and 512 for Zero. The diagnostic preserves all judged positives and samples other passages, reducing competition relative to the full corpus. Its search measurements and Table 2’s encoding sample describe separate query populations.

Summing these component medians and extrapolating to one million queries gives Nano and Zero time shares of 13% and 11% of Stella’s, respectively. Large savings versus the model’s own query encoder remain. At a fixed **ef** from 32 to 512, search medians differ by at most 7% across encoders; the added work comes from the larger settings required to reach their respective targets. These component-time shares measure neither sustained throughput nor equal absolute relevance.

The binary-quantized collection leaves a larger speed difference between Nano and Zero. This comparison uses encoding timings from the diagnostic’s own queries, measured separately from search. In an exploratory selection from the registered sweep, the fastest settings within 1% of each encoder’s own exact score give

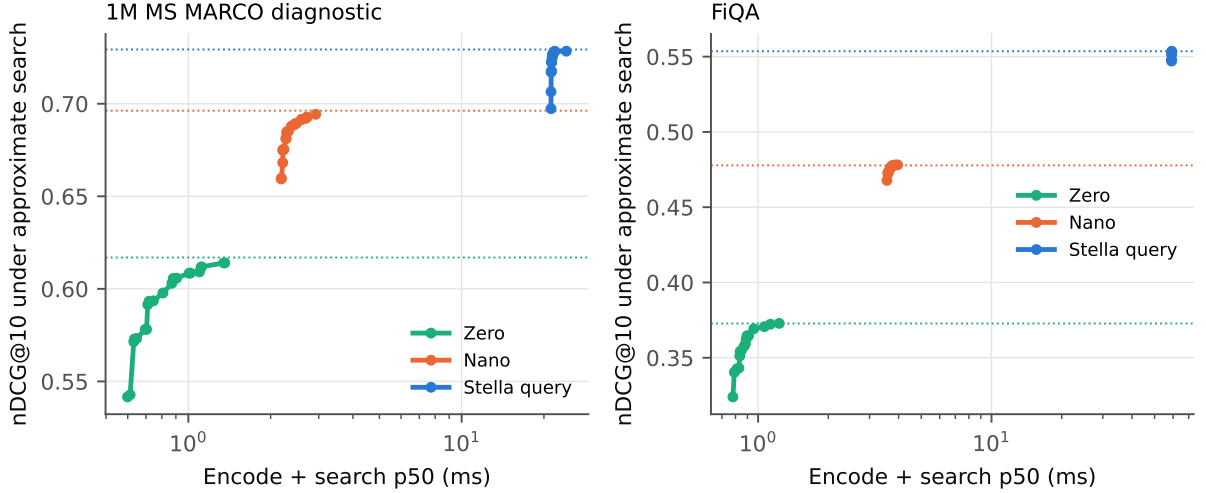


Figure 4: Approximate-search quality against encoding plus search time on FiQA and the positive-preserving MS MARCO diagnostic, registered sweep. Dotted lines are each encoder’s exact score.

encoding plus search medians of 21.3 ms for Stella, 2.48 ms for Nano, and 1.11 ms for Zero, at nDCG@10 of 0.7226, 0.6893, and 0.6116, respectively. These sums differ from the uncompressed extrapolated time shares above. Zero is 2.2-fold faster than Nano including search at these unequal quality levels.

**Construction and recurring cost.** Nano’s final optimization took a measured 57.3 A100 hours, about \$95 at the recorded rental rate. Zero’s retraining was historically estimated at 20 minutes with prepared targets, plus 8 to 12 hours of target encoding for a new embedding model. Preparation, data generation, recipe search, failed runs, evaluation, and engineering add to the full build. The four-to-seven-minute fitted screens answer a cheaper construction question than producing these optimized models. Once constructed, a compatible query encoder can amortize that one-time cost through recurring savings, provided its retained relevance meets the workload’s requirements.

A request-context policy can select among these choices. In an e-commerce example, Zero could serve searches while a user types, Nano after typing stops, and Stella on submission, with an `ef` setting for the chosen encoder accompanying each selection. This is an illustrative policy enabled by compatibility, not a measured typeahead workload. The policy assigns computation by interaction stage, so it needs no prediction of which encoder will retrieve better for a particular query.

Automatic relevance-based allocation is a separate problem. In the registered routing analysis over 12 datasets, a label-aware oracle selecting Zero or Nano scores 0.5473 against 0.5161 for always-Nano. Yet query length, subwords per word, and pooled-vector norm improve at most 0.005 over random routing at matched Nano share. The oracle shows the potential benefit of choosing between Zero and Nano for each query. The tested features capture little of that benefit. The oracle requires relevance information unavailable at request time.

## 7. Discussion

The findings separate two consequences of replacing a query encoder. Exact retrieval measures how well the replacement ranks the stored documents. Approximate search measures the additional loss introduced by searching their graph. A student can therefore reduce encoding cost while requiring greater graph-search breadth, and these effects must be considered together when interpreting the resulting savings.

These results have three consequences for a search system. First, a candidate query encoder is cheap to fit and to measure on development queries for the target index. On the original model list, that measurement selected better students than the model’s own retrieval score did. The vector-agreement counterexamples

show that cosine fidelity cannot certify the choice. Second, a replacement query encoder has its own search requirements. The distance ratio for the models’ own queries flags embedding models with larger recovery gaps before any replacement is fitted; the size of the penalty on a new workload still needs measurement on that workload. Third, a compatible family turns query compute into a per-request setting over the same stored documents. Allocation by request context needs no predictor; predicting which encoder will retrieve best for a given query remains unresolved.

The comparison with bge-small and LEAF concerns the value of retaining a shared index. bge-small scores 0.5171 and LEAF 0.5402 on BEIR-15, above Nano’s 0.5081, using their own document indexes. Keeping one Stella index lets each request choose among three query budgets; separate models would need separate indexes and ingestion pipelines to offer those choices.

## 8. Limitations

The quality study covers two closed-form recipes over related English embedding models at three dimensionalities. The shared vocabulary constrains the table’s model list; transfer to other tokenizers, vendor-provided embeddings, and fully trained students remains open. Dimensionality covaries with family and quality. Intervals from resampling the models describe this model list, while family-held-out and prospective tests probe specific extensions.

The search study covers one engine, one graph configuration, and three corpora, including TREC-COVID’s 50 queries. Its geometry predictor forecasts recovery gaps, not their mechanism. Timing across embedding models used a shared machine; latency conclusions use the separate Constella sweep. LightRetriever recurrence covers one released model and two corpora.

Constella’s training pools differ, including FEVER exposure; Stella discloses ArguAna, FiQA, and FEVER exposure. Quality intervals condition on fixed models. Timing covers warmed sequential CPU execution, while the million-passage diagnostic preserves positives. Request-context selection is a capability and an illustrative policy; concurrent operation and real timeahead quality were not measured.

## Artifact availability

The [companion repository](#) contains inference examples, pinned configurations, dated methods and amendments, immutable result receipts, and the full research history, including secondary analyses and negative results. Released models are [Constella Zero](#), [Constella Nano](#), and the [Stella document encoder](#). The three encoders are available through FastEmbed, Qdrant’s embedding library. The cleaned 337,981-query fit list and the one-million-passage identifier list are identified by hash but not yet archived. Prepared training corpora are described by source and file manifests.

## Appendix A. Registered evaluations

**Serving configuration.** Stella is pinned to revision `ffeb2b7e`, with its `s2p_query` prompt, fp32 encoding, normalized output, fp16 stored document vectors, and cosine similarity; maximum sequence length 512. Zero’s training pool includes FEVER-train and HotpotQA; Nano excludes FEVER. MS MARCO is licensed for non-commercial use, so it was validation-only in our distillation: it supplied no training targets, negatives, or generation seeds, although Nano’s pretrained base has prior exposure to it.

**Partitions.** The six-set suite is NFCorpus, SCIDOCs, SciFact, TREC-COVID, FiQA, and ArguAna. “Clean-four” means the first four, which have no disclosed Stella overlap; it is not a causal contamination estimate. Four datasets were held out and evaluated once: FEVER, DBpedia-entity, and the CQADupStack android and english forums. Two other forums, physics and programmers, select table penalties and routing settings and were reused during model construction.

**Nano’s registered contrasts** report nDCG@10 differences and used a fixed sequence with one-sided 2.5% lower bounds. “Established” means the bound cleared zero; “unresolved” means it did not; “descriptive” rows were registered as reports, not tests. Intervals resample queries with the models fixed.

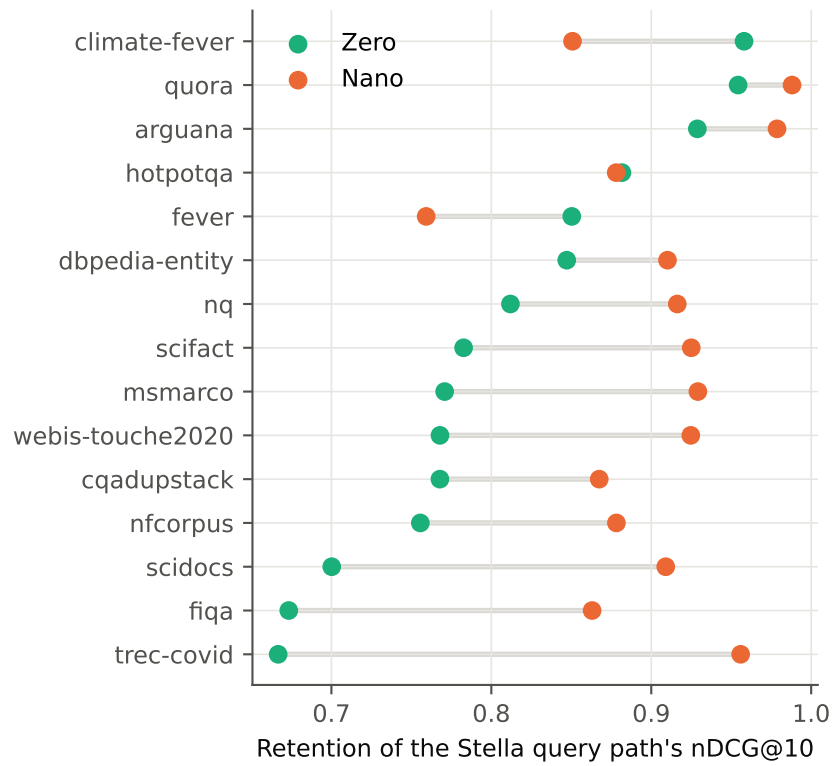


Figure A1: Retention per dataset. Datasets with disclosed Stella training exposure are ArguAna, FiQA, and FEVER; the four originally held-out sets are FEVER, DBpedia-entity, and the CQADupStack android and english forums.

| Contrast         | Partition                                             | Score difference             | Outcome     |
|------------------|-------------------------------------------------------|------------------------------|-------------|
| Nano – bge-small | clean-four                                            | +0.0176; lower bound +0.0037 | established |
| Nano – bge-small | all six                                               | +0.0274; lower bound +0.0173 | established |
| Nano – LEAF      | all six                                               | +0.0162; lower bound +0.0065 | established |
| Nano – LEAF      | clean-four                                            | −0.0011; lower bound −0.0145 | unresolved  |
| Nano – bge-small | held-out                                              | +0.0032 [−0.0069, +0.0134]   | descriptive |
|                  | non-double-exposure<br>aggregate, dataset<br>weighted |                              |             |
| Nano – bge-small | held-out                                              | −0.0121 [−0.0219, −0.0024]   | descriptive |
|                  | non-double-exposure<br>aggregate, query<br>pooled     |                              |             |
| Nano – LEAF      | held-out                                              | −0.0389 [−0.0488, −0.0290]   | descriptive |
|                  | non-double-exposure<br>aggregate                      |                              |             |

The registered non-double-exposure aggregate uses DBpedia-entity weighted one half, the android and english forums one quarter each, FEVER excluded as the double-exposure sensitivity row.

**Zero’s registered contrasts** report nDCG@10 differences and used a Holm correction across the family. The dense comparison tests whether Zero exceeds LightRetriever’s dense path under per-task prompts, whose six-set nDCG@10 is 0.4583. Zero scores below that reference and does not confirmatorily beat BM25.

| Contrast, all six                    | Score difference | Raw 95% interval   | Outcome                                 |
|--------------------------------------|------------------|--------------------|-----------------------------------------|
| Zero – LightRetriever<br>dense       | −0.0243          | [−0.0405, −0.0086] | below LightRetriever                    |
| Zero – BM25                          | +0.0165          | [+0.0017, +0.0311] | unresolved; p=.0149<br>fails Holm .0083 |
| Zero + BM25 (convex)<br>– OpenSearch | +0.0043          | [−0.0063, +0.0151] | unresolved                              |

The registered clean-four sensitivity is descriptive: −0.0443 [−0.0675, −0.0212], −0.0311 [−0.0517, −0.0109], and −0.0107 [−0.0262, +0.0043]. Unresolved superiority does not establish equivalence. The confirmatory fusion used convex score fusion with development-fitted weight 0.8 at depth 1000. Later distribution-based score fusion at 100 candidates is a deployability choice, not a new confirmation.

**The held-out four**, copied from their one-shot published aggregate (registered):

| System      | FEVER  | DBpedia | Android | English |
|-------------|--------|---------|---------|---------|
| Nano        | 0.6231 | 0.4190  | 0.4774  | 0.4298  |
| Zero        | 0.6978 | 0.3900  | 0.4431  | 0.3690  |
| Stella      | 0.8207 | 0.4603  | 0.5275  | 0.5183  |
| bge-small   | 0.8662 | 0.4003  | 0.4761  | 0.4558  |
| LEAF        | 0.8671 | 0.4520  | 0.5260  | 0.4707  |
| BM25        | 0.5036 | 0.3045  | 0.3952  | 0.3481  |
| Zero + BM25 | 0.7559 | 0.4030  | 0.4629  | 0.4016  |
| Nano + BM25 | 0.6879 | 0.4296  | 0.4750  | 0.4323  |

**Precision.** All released encoders run in fp32 through the published loader.

## Appendix B. Reproduction

**Closed-form table student.** Let  $X$  contain token counts divided by query token count,  $Y$  the original model’s own query vectors, and  $W_0$  its normalized outputs for individual tokens encoded between special tokens. Fit

$$\min_W \|XW - Y\|_F^2 + \lambda \|W - W_0\|_F^2.$$

The anchor includes the original model’s pooling and output projection. Student tokenization includes special tokens, has no query prefix, and truncates at 512 tokens; mean pooling and L2 normalization follow lookup. Block conjugate gradient solves the normal equations, requiring every output column to pass relative-residual tolerance 1e-6 in fp32 or 1e-8 in fp64. Penalties range from 1e-4 to 1e-1, with one decade extension when development selection lands at an edge. Exact development nDCG@10 selects the penalty before six-set scoring. The fit list was cleaned after an earlier version had 1.31% held-out-query overlap. One checkpoint in the expanded model list, arctic-embed-m-v1, stopped at the convergence gate and is excluded from the 26. Encoder exposure was documented for the registered model list; the exploratory expansion was not individually audited.

**Closed-form transformer student.** Frozen bge-small revision 5c38ec7c supplies masked-mean states from layers 12, 8, and 4. Concatenating a constant bias feature yields  $F$  with 1,153 columns. A dense solve fits

$$\min_A \|FA - Y\|_F^2 + \lambda \frac{\text{tr}(F^\top F)}{1153} \|A\|_F^2.$$

The bias is regularized; predictions are normalized. Penalty selection matches the table. bge-small is itself an entry in the model list sharing this backbone. The bge-small transformer student retains 1.008 of bge-small’s own retrieval score; the gte-small transformer student, whose model is nearly a linear image of the backbone, retains 0.978 of gte-small’s own score. Removing bge-small changes the selection score gaps reported in Section 4 by less than 0.06. The original model’s query prompts apply to targets, while students have no query prefix.

**Zero: data and distillation.** The pool has 338,076 usable query-document pairs and 220,632 query-only rows from Amazon ESCI, FEVER-train, HotpotQA-train, SQuAD-train, Mr. TyDi English, NQ Open, and TriviaQA. Training sources permit commercial derived weights, with attribution where required. The distillation phase also uses 924,704 short document spans as pseudo-queries. Table rows start from Stella’s normalized token-in-context outputs; scalar weights start from inverse document frequency. Stella targets and document vectors stay frozen.

The first phase runs 16,000 steps, batch 512, with ordinary count pooling. Half the batch comes from query-only rows and pseudo-queries with cosine alignment loss. Paired queries combine cosine alignment and KL

divergence against Stella’s score distribution over a positive and 31 distractors from a document bank; the extra-text cosine mean is added separately. Cosine and KL weights are 1, temperature 0.02. Adam learning rates are 3e-3 for rows and 1e-2 for weights. A row penalty of 1e-3 anchors initialization, divided by one plus the row’s update count.

**Zero: ranking and export.** The second phase runs 2,500 contrastive steps from that checkpoint. InfoNCE uses 32,768 bank negatives plus provided negatives, at temperature 0.02. Known positives and candidates scoring above the positive’s Stella score minus 0.02 are masked. Adam peaks at 1e-3 for rows and 1e-2 for weights, with 200 warmup steps and linear decay to a tenth. The row penalty anchors the phase’s starting checkpoint. Both phases use seed 0 and a bank of 2,000,000 document vectors.

Development selected the two-phase recipe, then square-root count pooling for its unchanged table. Export folds scalar weights into rows and applies per-row absolute-maximum int8 quantization with fp32 scales. A token appearing  $c$  times contributes  $\sqrt{c}$  times its dequantized row before normalization. Unupdated rows retain initialization. Development comprises NQ, HotpotQA, two CQADupStack forums, and two training-source slices..

**Nano: data and initialization.** The deduplicated pool contains 3,486,034 query texts: 403,443 real queries from ESCI, HotpotQA, Mr. TyDi, NQ Open, SQuAD, and TriviaQA; 999,744 PAQ questions; 1,248,385 harvested texts; and 834,462 generated queries. Harvested text comes from Wikipedia, arXiv metadata, and the licensed document pool. Sampling gives equal presentation shares to 12 forms: factoid, how-to, claim, argument, finance, title, keyword, health, product, comparison, yes/no, and conversational. The 6,070,049 eligible documents replay in reshuffled epochs. FEVER is excluded. Queries have no student prefix; document examples use `passage::`; Stella supplies corresponding query or document targets.

Qwen3-8B-AWQ revision 4da05a8e generates seven forms with thinking disabled, temperature 0.8, top-p 0.95, and passage-derived seeds. Finance and health use Wikipedia passages; other forms use HotpotQA. Filtering applies length checks, exact and near-duplicate handling, document hold-outs, overlap screens, and copied-span rejection. Pinned prompts, sampling limits, filters, and source manifests remain in the companion repository.

The bge-small backbone uses revision 5c38ec7c. Its three-layer features project to 1,024 dimensions, then masked mean pooling and normalization. Ridge initialization uses 60,000 queries, seed 21, and a 50,000/10,000 split to select normalized squared-L2 error over penalties 1e-6 to 1; the selected 0.001 penalty is refitted on all queries. The affine projection commutes with mean pooling.

**Nano: optimization and selection.** Backbone and projection train against normalized Stella vectors with squared-L2 loss, alternating three query batches and one document batch. AdamW uses batch 32, seed 0, betas (0.9, 0.999), epsilon 1e-8, matrix weight decay 0.01, zero decay for one-dimensional parameters, and gradient clipping at norm 1. Forward passes use bf16; loss uses fp32. Three cycles decay from 1e-4 to 1e-5, with 64,000-example warmup in the first. The released third cycle-end checkpoint follows 199,999,721 example presentations. Registered screening and monitoring use an equally weighted four-family development macro: six BRIGHT slices, MedicalQARetrieval, LEDGER, and LegalBench CorporateLobbying and ConsumerContractsQA. Defaults remain when comparisons are unresolved. Full configuration, generation prompts, selection rules, and artifact hashes are retained in the repository.

**Search protocol.** Qdrant 1.19.1 uses cosine HNSW with  $m=16$ ,  $ef\_construct=100$ , indexing threshold 1 KB, and  $ef$  from 16 to 512. Graphs for the comparison across embedding models cover FiQA, SCIDOCS (25,657 documents), and TREC-COVID (171,332), uncompressed, with two seeded insertion-order builds per workload. Search requests 11 results before the registered self-hit drop. Recovery counts a returned document when its exact score reaches the exact 10th-neighbor score within cosine tolerance 1e-4. NumPy parity is checked on 200 queries per encoder; embedding models below 0.995 are excluded. This removes tas-b, whose SCIDOCS transformer student parity is 0.994. The arctic-embed-l mean-pooling control is outside the model-list statistics.

The smallest student  $ef$  reaching the relative nDCG loss of the original model’s own query encoder at  $ef=64$ , divided by 64, is computed per build and averaged. Recovery uses the analogous rule. Failure by  $ef=512$  is censored, with multiplier at least eight. SCIDOCS addition, tie handling, parity threshold, and exclusion

amendments preceded analysis of results for any embedding model; dated methods retain their history. Gap-predictor intervals resample each embedding model with all its workload results. The distance ratio uses a 5,000-document neighbor sample; family-held-out ridge prediction uses standardized features and unit penalty.

LightRetriever replication uses the released Qwen2.5-1.5B adapter and primary web-search instruction, comparing lookup and full query encoding over the same document vectors under the search protocol for the comparison across embedding models. The repository pins its revision and prompt.

**Instance and timing workloads.** FiQA uses all 57,638 documents and 648 queries. The MS MARCO diagnostic keeps every judged-positive passage for 6,980 development queries, then uniformly samples without replacement to 1,000,000 passages from the 8,841,823-passage corpus, seed 20260930. MS MARCO is validation-only, supplying no distillation targets, negatives, or generation seeds; Nano’s pretrained backbone has prior exposure.

The registered instance sweep checks exact parity against NumPy and measures warmed encoding and search separately, sequentially. Compressed sweeps use oversampling one, two, and four. The separate encoding sample contains 100 five-to-12-word queries, batch one, four CPU threads, Apple M5 Pro.

## Appendix C. Embedding models and prospective scores

**Original model list.** All scores are six-set exact nDCG@10. Pool is document-vector pooling: cls denotes the classification-token representation, and mean denotes mean pooling. Reg. marks the ten registered embedding models; Expl. marks the exploratory expansion. Pinned revisions and prompts are in the companion configurations.

| Encoder             | Dim. | Pool | Original | Table student | Transformer student | Status |
|---------------------|------|------|----------|---------------|---------------------|--------|
| bge-small-en-v1.5   | 384  | cls  | 0.5042   | 0.3581        | 0.5084              | Expl.  |
| arctic-embed-s      | 384  | cls  | 0.4993   | 0.3516        | 0.2594              | Expl.  |
| gte-small           | 384  | mean | 0.4838   | 0.2903        | 0.4733              | Expl.  |
| arctic-embed-xs     | 384  | cls  | 0.4662   | 0.3440        | 0.2779              | Expl.  |
| e5-small-v2         | 384  | mean | 0.4544   | 0.3203        | 0.3367              | Expl.  |
| minilm-l12          | 384  | mean | 0.4219   | 0.3138        | 0.3229              | Expl.  |
| minilm-l6           | 384  | mean | 0.4142   | 0.3267        | 0.3088              | Expl.  |
| multi-qa-minilm-l6  | 384  | mean | 0.4008   | 0.3406        | 0.3094              | Expl.  |
| msmarco-minilm-l6   | 384  | mean | 0.3281   | 0.2905        | 0.1962              | Expl.  |
| gte-base-en-v1.5    | 768  | cls  | 0.5331   | 0.3252        | 0.2618              | Reg.   |
| arctic-embed-m-v1.5 | 768  | cls  | 0.5263   | 0.3279        | 0.1874              | Reg.   |
| bge-base-en-v1.5    | 768  | cls  | 0.5259   | 0.3529        | 0.3249              | Reg.   |
| bge-base-en-v1      | 768  | cls  | 0.5131   | 0.3004        | 0.2773              | Expl.  |
| gte-base            | 768  | mean | 0.5063   | 0.2762        | 0.2314              | Expl.  |
| e5-base-v1          | 768  | mean | 0.4934   | 0.3126        | 0.2345              | Expl.  |
| e5-base-v2          | 768  | mean | 0.4669   | 0.2930        | 0.2241              | Reg.   |

| Encoder              | Dim. | Pool | Original | Table student | Transformer student | Status |
|----------------------|------|------|----------|---------------|---------------------|--------|
| contriever-msmarco   | 768  | mean | 0.3909   | 0.3270        | 0.1879              | Expl.  |
| tas-b                | 768  | cls  | 0.3262   | 0.2115        | 0.0868              | Expl.  |
| contriever           | 768  | mean | 0.2846   | 0.1864        | 0.0590              | Expl.  |
| gte-large-en-v1.5    | 1024 | cls  | 0.5970   | 0.2455        | 0.2229              | Reg.   |
| stella-400M-v5       | 1024 | mean | 0.5745   | 0.3974        | 0.2505              | Reg.   |
| mxbai-embed-large-v1 | 1024 | cls  | 0.5368   | 0.2605        | 0.2641              | Reg.   |
| bge-large-en-v1.5    | 1024 | cls  | 0.5329   | 0.2845        | 0.2679              | Reg.   |
| arctic-embed-l       | 1024 | cls  | 0.5289   | 0.3034        | 0.1564              | Reg.   |
| gte-large            | 1024 | mean | 0.5129   | 0.2481        | 0.2249              | Expl.  |
| e5-large-v2          | 1024 | mean | 0.4735   | 0.2585        | 0.1743              | Reg.   |

**Prospective model list (exploratory).** Eight encoders were fixed before encoding. Predictions from the original 26 were committed by hash before student fitting, with each command verifying that hash. The e5-base-unsupervised table stopped at the convergence gate at penalty 1e-5; its completed transformer student remains included. Each student column pairs actual and predicted six-set nDCG@10.

| Encoder                    | Dim. | Original | Table student, actual / predicted | Transformer student, actual / predicted |
|----------------------------|------|----------|-----------------------------------|-----------------------------------------|
| bge-large-en v1            | 1024 | 0.5237   | 0.2613 / 0.2883                   | 0.2481 / 0.2103                         |
| e5-large                   | 1024 | 0.4914   | 0.2875 / 0.2757                   | 0.2243 / 0.1825                         |
| nomic-embed-text-v1        | 768  | 0.4851   | 0.3684 / 0.2945                   | 0.2418 / 0.2336                         |
| e5-base-unsupervised       | 768  | 0.4562   | stopped / 0.2833                  | 0.1178 / 0.2087                         |
| all-MiniLM-L12-v1          | 384  | 0.4115   | 0.3256 / 0.3171                   | 0.3340 / 0.3062                         |
| msmarco-distilbert-base-v4 | 768  | 0.3436   | 0.3071 / 0.2395                   | 0.1734 / 0.1117                         |
| msmarco-bert-base-dot-v5   | 768  | 0.3239   | 0.2127 / 0.2318                   | 0.1032 / 0.0947                         |
| paraphrase-MiniLM-L6-v2    | 384  | 0.3001   | 0.2566 / 0.2738                   | 0.2447 / 0.2103                         |

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